

PROJECT DATA SHEET



CASE Consultants

Civil And Structural Engineers

Moose Hall

Barewell Road

Torquay

TQ1 4PA

tel. (01803) 200175

fax. (01803) 323205

E-Mail: admin@caseconsultants.net

JOB/PROJECT : NOSS ON DART MARINA, DARTMOUTH

JOB/PROJECT NO : C8829 SHEET NO : COVER

GENERAL :

CLIENT : PREMIER MARINAS

ARCHITECT : N/A

AUTHORITY : TORBAY COUNCIL

JOB/PROJECT DESCRIPTION : CANTILEVERED PLATFORM SCAFFOLD

PROPOSED USE: ROOF ACCESS PLATFORM

OVERALL SIZE : AS CALCS

No. OF STOREYS : N/A

TYPE OF SOIL INVESTIGATION : N/A

BEARING STRATA : LEVEL 5 SLAB

TYPE OF FOUNDATION : CONCRETE SLAB

MAXIMUM SAFE BEARING PRESSURE

N/A kN/m²

STRUCTURAL MATERIALS USED : STANDARD SCAFFOLD COMPONENTS

STANDARDS AND CODES USED IN DESIGN :

General actions:
Densities, self weight &
imposed loads

BS EN 1991-1-1



Steelwork

BS EN 1993 (EC3)



Timber

BS EN 1995 (EC5)



Masonry

BS EN 1996 (EC6)



General actions:
Actions on structures
exposed to fire

BS EN 1991-1-2



Others

TG20: 21
Technical guidance



Snow Loads

BS EN 1991-1-3



Wind Loads

BS EN 1991-1-4



Concrete

BS EN 1992 (EC2)



AUTHORISATION OF CALCULATION SHEETS AND BENDING SCHEDULES :

CALCULATION SHEET NOS.	CHECKED BY	DATE
Sheet nos: CP01 - CP05 inclusive	YC /hw	18/9/25



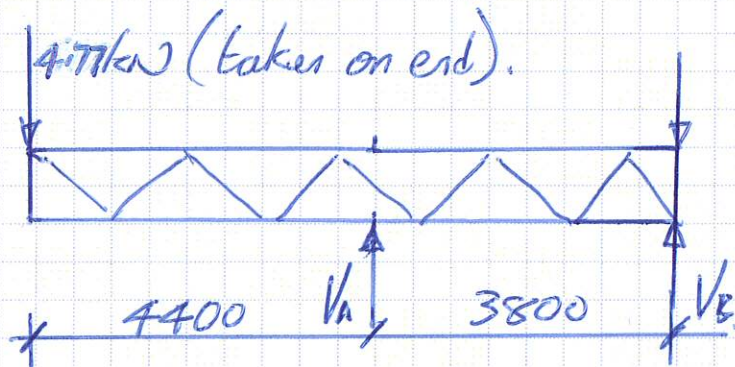
Calculations

Contract	<u>NOSS ON DART MARINA</u> <u>DARTMOUTH</u>	Job No.	<u>C8829</u>	Sheet No.	<u>CP 1.</u>
		Designed by	<u>A.C.H.</u>	Date:	<u>SEPT 25</u>

These calculations cover a design assessment of the cantilevered roof platform shown on optima drawing No. 25/OPT/21740-002

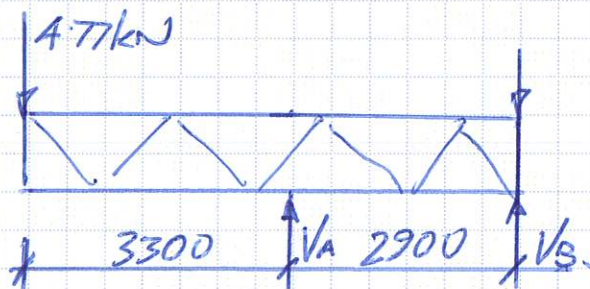
From Optima design self weight & imposed load for 2.0m bay for the scaffold = 4.77kN/bay. (Char).

From layout drawing the critical design case occurs at the corner where the cantilever distance is greatest and the splay concentrates the uplift on the backspan. Consider 4.4m cantilever.



$$V_B = 5.44 \text{ kN, uplift} \quad \& \quad V_A = 10.14 \text{ kN.}$$

Consider 3.3m cantilever



$$V_A = 10.0 \text{ kN} \quad \& \quad V_B = 5.35 \text{ kN.}$$

These are similar to Optima values of $V_A = 12 \text{ kN}$ & $V_B = 6.0 \text{ kN}$.



Calculations

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CP2.

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A.C.H.

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SEPT 25

Consider deflection, for 4.4m cantilever.

For 780mm alloy beam. $E = 70,000 \text{ N/mm}^2$
 $I = 16500 \text{ cm}^4$

$$\therefore EI = 1.155 \times 10^3 \text{ N/mm}^2$$

$$\delta = \frac{Pa^2(L+a)}{3EI} = \frac{4.77 \times 10^3 \times 4400^2 (3800 + 4400)}{3 \times 1.155 \times 10^3}$$

$$= 21.5 \text{ mm} = \text{span} / 204 < \text{span} / 180$$

Optima Value 11.72 mm. $\nearrow$

$\therefore \text{ok.}$

Consider deflection for 3.3m cantilever.

$$\delta = \frac{4.77 \times 10^3 \times 3300^2 (3300 + 2900)}{3 \times 1.155 \times 10^3}$$

$$= 9.29 \text{ mm} = \text{span} / 355 < \text{span} / 180 \therefore \text{ok.}$$

$\nearrow$ Optima Value 4.94 mm.

Consider kentledge.

-worst case occurs where 3 No. double standards occur on the corner.

Total uplift = $6 \times 6.0 \text{ kN} = 36 \text{ kN}$, over a 5.0m length.

Allowing 1.5 F.O.S gives uplift = 54 kN.

for 275 thk slab characteristic resistance = 6.6 kN.

$$\therefore \text{width of slab reqd} = 54 / (6.6 \times 5.0) = 1.60 \text{ m.}$$



Calculations

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Line of kentledge standards is 600mm from the edge of the roof slab.

The spread of the point load will be $\frac{8400}{(2 \times 3)}$
= 1400mm

∴ notional spread of line load = $600 + 1400$
= 2000mm > 1600
∴ ok.

Consider support to outer standards. These are to be dead shored down to ground floor slab level.
- No further consideration required.

∴ EXISTING BUILDING IS SATISFACTORY
TO SUPPORT THE SCAFFOLD REACTIONS.



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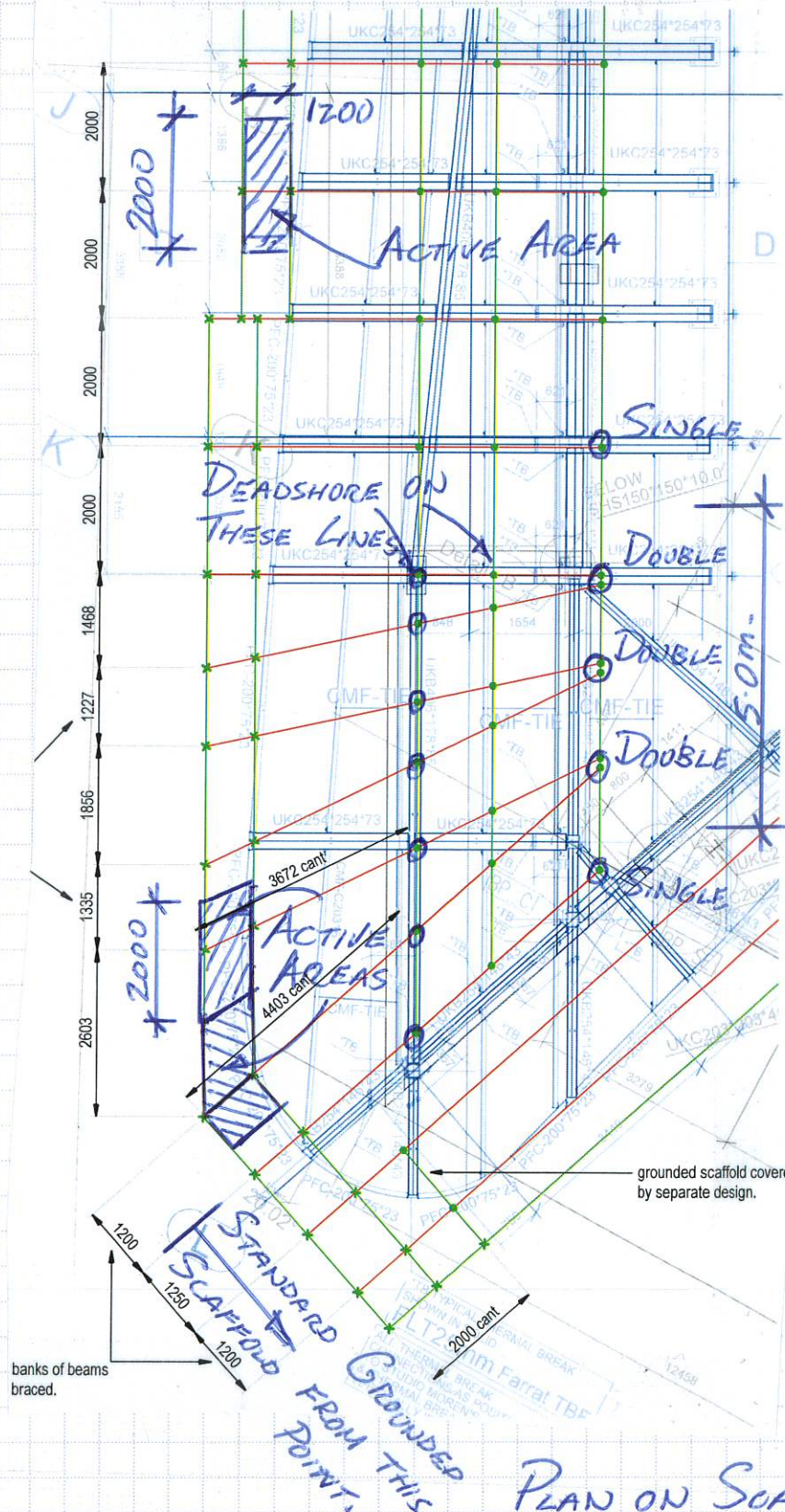
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Calculations

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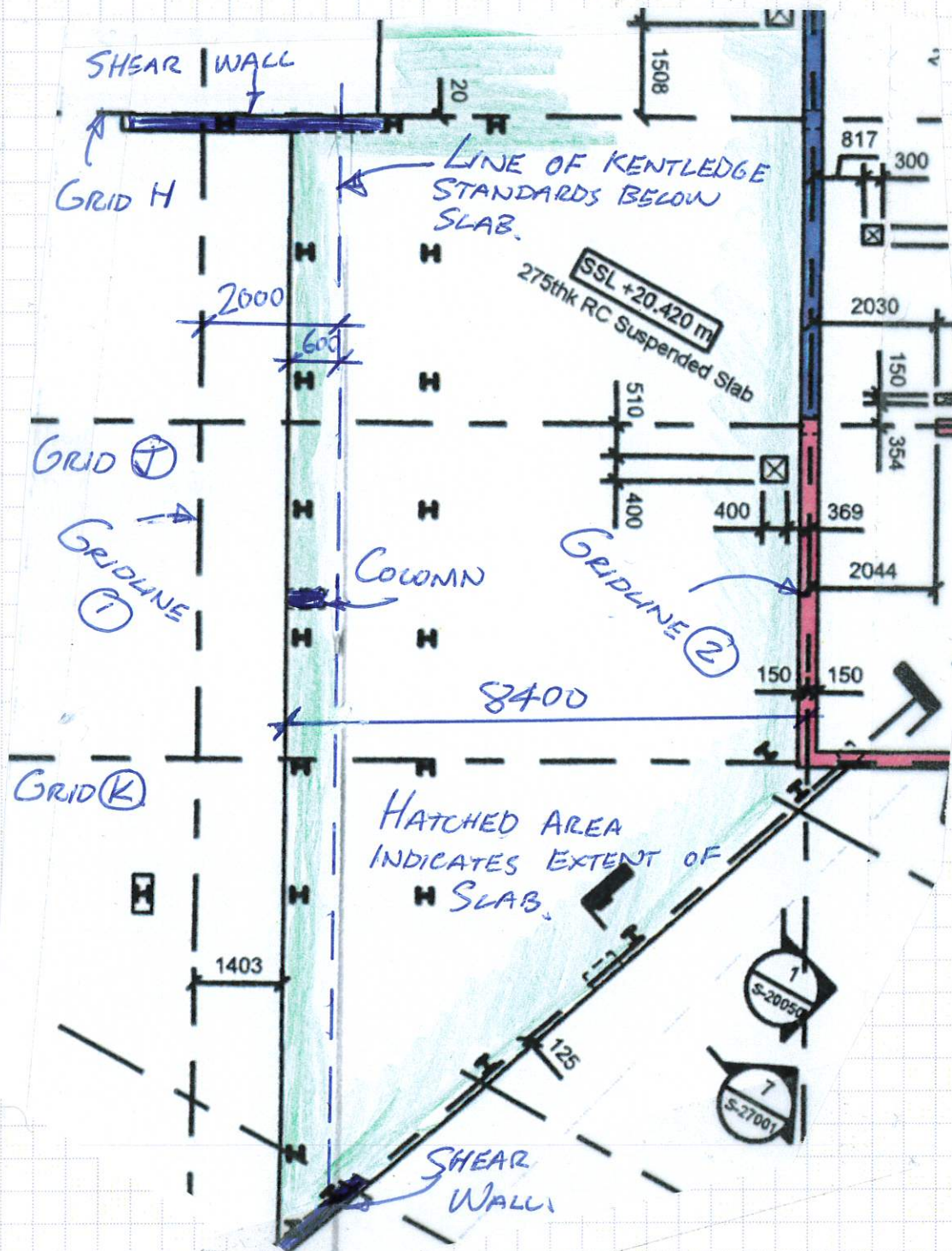
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ROOF SLAB GA.